

VINCENT XAYASAK

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ML engineer skilled in LLM-powered system design, production RAG deployment, multimodal vision-language models, large-scale evaluation benchmark construction, and rigorous model evaluation; experienced bridging research and production to improve software quality and reliability.

EDUCATION

University of California, Davis | Davis, CA
B.S. in Computer Science | GPA: 3.56

June 2026

Relevant Coursework: Machine Learning, Software Engineering (CI/CD, unit testing, software quality), Computer Security & Privacy

PROFESSIONAL EXPERIENCE

Advantest | San Jose, CA
AI/ML Software Engineer Intern

June 2025 – September 2025

- **Fine-tuned Llama 3.3 70B via QLoRA** for SmarTest8 syntax; rapidly prototyped multiple algorithmic approaches, scaling from a 4GB proof-of-concept to a 48GB production deployment.
- **Engineered a production RAG pipeline** with BAAI/bge-m3 embeddings to eliminate hallucinations, improve output quality, and accelerate retrieval across large-scale structured documentation.
- **Automated firmware validation and failure analysis workflows** via LLM-powered CI/CD-integrated pipelines, reducing engineering turnaround from one week to 5 hours (~90% efficiency gain).

RESEARCH

BRIDGE: Bias Register-Integrated Dataset for LLM Bias Evaluation | Davis, CA
Research Assistant

January 2026 – May 2026

- **Designed an end-to-end data collection and labeling pipeline** producing a 19,421-sentence benchmark from seven corpora; ensured annotation quality under noisy, ambiguous ground-truth conditions via expert-adjudicated verification.
- **Designed GRDC**, a five-metric evaluation framework for register-aware LLM bias detection across eight frontier models; identified the Register Polarity Flip failure mode ($DC=1.0$, $p \approx 6 \times 10^{-5}$).
- **Co-authored paper in submission** to NeurIPS 2026 Evaluations & Datasets Track on register-stratified bias evaluation for institutional LLM auditing and software quality workflows.

Measuring the Shape of Reasoning-Model Behavior | Davis, CA
Research Assistant

January 2026 – May 2026

- **Designed Reasoning Path Consistency (RPC)**, a three-axis evaluation framework (lexical, structural, semantic) for systematic LLM output quality assessment, validated against human judgments (Spearman $\rho=0.34$).
- **Developed Parametric Sampling Fingerprints (PSF)** fitting Pass@k trajectories (mean $R^2=0.96$); evaluated 16 reasoning models, demonstrating RPC and PSF capture independent quality dimensions that scalar accuracy conceals.
- **Co-authored paper in submission** to ACL on multi-dimensional LLM capability measurement beyond single-number benchmark accuracy.

PROJECTS

Facial Acne AI | San Jose, CA
Personal Project — Multimodal Vision-Language System

April 2026 – June 2026

- **Built a multimodal VLM pipeline** using Qwen2.5-VL 7B to analyze facial images and produce structured observations; integrated vision inference with RAG over ingested dermatology literature for research-grounded answers.
- **Designed and evaluated a retrieval system** using ChromaDB and BGE embeddings; measured citation accuracy and answer quality across baseline vs. RAG-augmented conditions via a structured evaluation suite.
- **Deployed a full-stack Flask web application** with photo upload, multimodal inference, and configurable RAG toggle running entirely locally via LM Studio.

SKILLS & INTERESTS

Languages: Python, Java, C/C++, Go, SQL, JavaScript, HTML/CSS

AI/ML: PyTorch, Hugging Face, NLP, LLM Fine-tuning (LoRA/QLoRA), RAG, PEFT, ChromaDB, Vision-Language Models (VLM)

Evaluation & Data: Evaluation Framework Design, Benchmark Construction, Data Collection & Labeling Pipelines, Retrieval Systems, Semi-Structured Data Processing

ML Research: RL (GRPO/PPO), On-Policy Distillation, Bias Evaluation, LLM Output Quality Assessment, Adversarial Benchmarking

Dev & MLOps: Docker, Kubernetes, FastAPI, Flask, AWS, Git, pytest, CI/CD Pipelines (GitHub Actions), Cursor, Claude Code

Inference & Optimization: CUDA, Quantization (GGUF/AWQ), Concurrent Inference Pipelines

Interests: LLM Evaluation, Software Quality Systems, Multimodal ML, Model Reasoning & Bias, MLOps, NLP